

Keil Setup FAQ

Introduction

This document outlines common issues and solutions encountered while setting up a project in the Keil IDE, specifically when integrating STM32CubeMX and dealing with compiler and debugger configurations.

Q: What should I do if I encounter errors related to ARM Compiler version?

A common error is: *“Target ‘Project’ uses ARM-Compiler ‘Default Compiler Version 5’ which is not available.”* This occurs because STM32CubeMX defaults to generating code for ARM Compiler Version 5, while Keil may expect Version 6.

- **Solution:** In the project options, ensure that ARM Compiler Version 6 is selected. You can adjust this under **Options for Target -> Target Tab**.
- If needed, reinstall or update ARM Compiler packs using the Pack Installer, ensuring both Version 5 and 6 are available.

Q: What do I do if my project cannot find necessary DLLs? (e.g., SarmCM3.dll, STLinkUSB-Driver.dll)

- **SarmCM3.dll Error:** If Keil cannot locate **SarmCM3.dll** or similar files, ensure that the full path to these files is specified in **Options for Target -> Debug Tab**. If not, add it to the PATH environment variable.
- **STLinkUSBDriver.dll Error:** For ST-Link debugging errors related to missing DLLs (e.g., **ST-LINKIII-KEIL.SW0.dll**), ensure that the ST-Link USB drivers are installed, and the path to **STLinkUSBDriver.dll** is correctly set.

Q: How do I solve the error “Registered ARM Compiler version not found in path” even though it builds?

- This can be resolved by ensuring that the correct path to the ARM Compiler is set in the environment variable `PATH` or configured in Keil settings. The compiler path should point to `C:\Keil\ARM\BIN`.
- In **Options for Target** -> **Target Tab**, ensure that the correct compiler path is listed. ARM Compiler 6.22 or ARMCLANG is typically expected.

Q: What steps should I follow if Keil can’t find ST-LINKIII-KEIL.SWO.dll?

- Ensure that you have the ST-Link drivers installed. These can be found under the Keil installation folder, typically at `C:\Keil\ARM\STLink`.
- Make sure the correct path is referenced in **Options for Target** -> **Debug Tab**. You can add the full path to the `ST-LINKIII-KEIL.SWO.dll`.
- If ST-Link is not detected, switch to **ULINK/ME** for debugging, which sometimes resolves this issue.

Q: Why am I getting “Additional software components required” errors in Keil RTE?

This usually happens when using STM32CubeMX to generate code and Keil expects other components to be available.

- **Solution:** Use the Keil Pack Installer to ensure that all required packs are installed. The error message typically indicates what is missing.
- Check the **Manage Run-Time Environment (RTE)** for missing components, and use the **Pack Installer** to download them.

Q: What should I select in the RTE setup for a basic STM32CubeMX project?

Here are typical selections for a basic project:

- **CMSIS Core:** Include the CMSIS core components.

- **Device Startup:** Choose the appropriate startup files for your STM32 device.
- **STM32Cube Framework:** For a HAL-based project, select STM32Cube HAL for peripheral initialization.
- **Debugger:** Choose ST-Link for debugging (unless you're using another debugger like ULINK).

Q: How do I make semihosting work with Keil?

Keil supports semihosting, but ensure the following:

- Semihosting must be enabled via the project options (select `use semihosting`).
- Use appropriate `_write()` and `_read()` implementations for redirecting the standard output to the debugger's console.

Q: How do I add STM32CubeMX code generation to Keil projects?

If STM32CubeMX fails to integrate properly:

- First, generate the code using STM32CubeMX, then import it manually into Keil by copying the files over.
- Ensure that the include paths and all project settings match the Keil project environment.

Q: How do I reset Keil if it becomes unstable or can't find files?

- Resetting your Keil installation and reinstalling the packs can often resolve issues with missing files or paths.
- Check all environment variables and make sure the Keil paths are set correctly.
- If all else fails, re-install Keil, ensuring that all needed software packages are installed via the Pack Installer.

Q: What if Keil can't save files or throws "Member not found" errors?

This might be due to a corrupted session or unsaved changes. If Keil locks up and throws errors such as `HRESULT DISP_E_MEMBERNOTFOUND`, try the following:

- Close Keil and re-open the project.
- If the issue persists, reset your project or open a backup.

Conclusion

These are the main issues and resolutions when setting up a Keil project for STM32 microcontrollers. With the right settings and configurations, Keil can be a powerful and stable tool for embedded development.